

SA-MIRI 2025

Practice Pg: Parallel

Training of Neural Networks

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Task 10.1 Task Setup and File Structure

- CIFAR-10 is pre-loaded in a shared MN5 path:
/gpfs/projects/nct_345/cifar-utils/cifar-10-batches-py
- `load_cifar` returns two `PrefetchDataset` without errors

```
[nct01029@allogin1 ~]$ ls /gpfs/projects/nct_345  
MN5-NGC-TensorFlow-23.03.sif  cifar-utils  cifar-utils.zip  secret.txt
```

```
/Chapter10  
├── ResNet50_seq.py  
├── ResNet50_seq.CPU.slurm  
├── ResNet50_seq.GPU.slurm  
├── ResNet50.py  
├── ResNet50.slurm  
└── cifar-utils  
    └── cifar.py
```

Task 10.2 Code Review and Understanding

- Train size

```
parser = argparse.ArgumentParser()
parser.add_argument('--epochs', type=int, default=5)
parser.add_argument('--batch_size', type=int, default=2048)
```

- Load dataset + model

```
train_ds, test_ds = load_cifar(batch_size)

model = tf.keras.applications.resnet_v2.ResNet50V2(include_top=True, weights=None,
                                                    input_shape=(128, 128, 3), classes=10)
```

- Train model on dataset

```
model.compile(loss='sparse_categorical_crossentropy',
              optimizer=opt,
              metrics=['accuracy'])

model.fit(train_ds, epochs=epochs, verbose=2)
```

Task 10.3 Training on CPU (Baseline Performance)

```
[nct01042@alogin1 Chapter.10]$ cat ResNet50_seq_CPU.31657392.out
```

```
Job Name: ResNet50_seq_CPU
```

```
Job ID: 31657392
```

```
Submit Directory: /gpfs/home/nct/nct01042/Chapter.10
```

```
Partition: acc
```

```
Account: nct_345
```

```
Number of Nodes Allocated: 1 (as02r1b05)
```

```
Total number of tasks: 1
```

```
Number of tasks per node: 1
```

```
Number of CPU cores per task: 20
```

```
Epoch 1/2
```

```
196/196 - 223s - loss: 2.0283 - accuracy: 0.2554 - 223s/epoch - 1s/step
```

```
Epoch 2/2
```

```
196/196 - 216s - loss: 1.7143 - accuracy: 0.3700 - 216s/epoch - 1s/step
```

Task 10.4 GPU Execution Time and CPU vs GPU Comparison

- CPU -> 223/216
- GPU -> 23/14
- CPU $\approx 10 \times$ GPU

```
[nct01042@login1 Chapter.10]$ cat ResNet50_seq_GPU.31657588.out
```

```
Job Name: ResNet50_seq_GPU
Job ID: 31657588
Submit Directory: /gpfs/home/nct/nct01042/Chapter.10
Partition: acc
Account: nct_345
```

```
Number of Nodes Allocated: 1 (as01r3b15)
Total number of tasks: 1
Number of tasks per node: 1
Number of CPU cores per task: 20
```

```
Specific GPUs allocated:
Fri Nov 7 11:12:47 2025
```

NVIDIA-SMI 535.86.10			Driver Version: 535.86.10		CUDA Version: 12.2		
GPU	Name	Perf	Persistence-M	Bus-Id	Disp.A	Volatile Uncorr. ECC	
Fan	Temp		Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M.
						MIG M.	
0	NVIDIA H100		On	00000000:9D:00:00	Off		0
N/A	39C	P0	61W / 700W	2MiB / 65247MiB		0%	Default Disabled

Processes:							
GPU	GI	CI	PID	Type	Process name	GPU Memory	
	ID	ID				Usage	
No running processes found							

```
Epoch 1/2
```

```
196/196 - 23s - loss: 2.0482 - accuracy: 0.2478 - 23s/epoch - 120ms/step
```

```
Epoch 2/2
```

```
196/196 - 14s - loss: 1.7463 - accuracy: 0.3632 - 14s/epoch - 73ms/step
```

Task 10.6 Analyze the Impact of GPU Parallelism

- First epoch slow down
- Second epoch speedup

```
Model ResNet50 1 GPUs
-----
Batch_size: 2048
Num_replicas: 1

Epoch 1/2
25/25 - 29s - loss: 2.2371 - accuracy: 0.1710 - 29s/epoch - 1s/step
Epoch 2/2
25/25 - 13s - loss: 2.1333 - accuracy: 0.2192 - 13s/epoch - 539ms/step

Model ResNet50 2 GPUs
-----
Batch_size: 4096
Num_replicas: 2

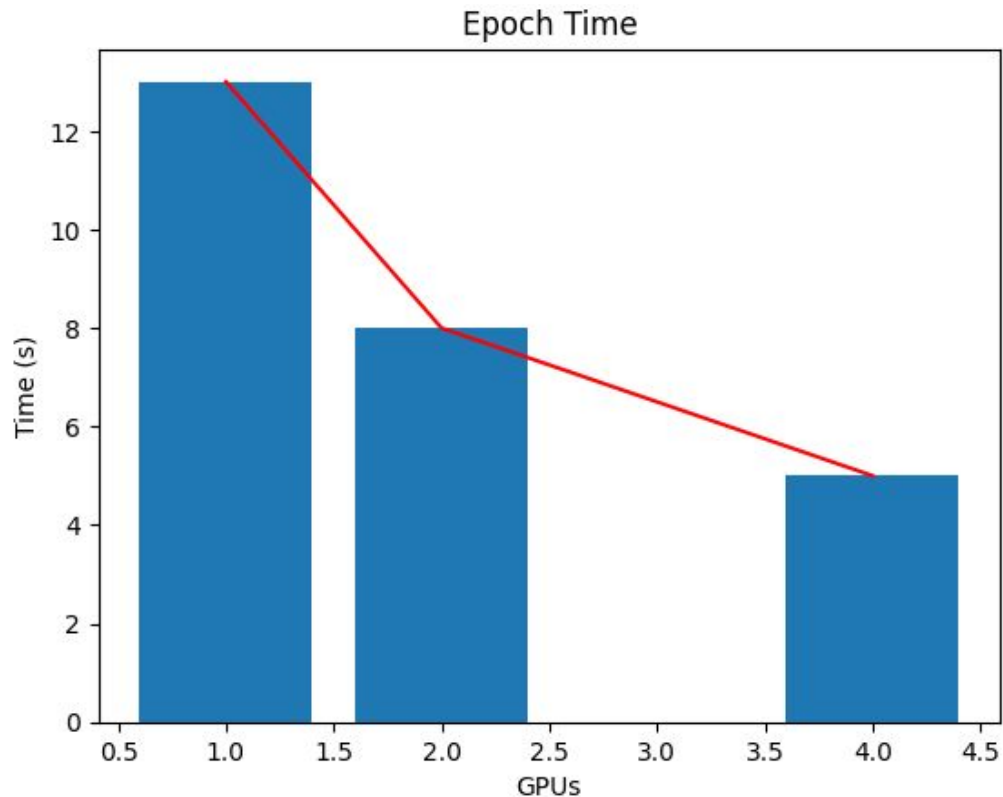
Epoch 1/2
13/13 - 31s - loss: 2.2839 - accuracy: 0.1477 - 31s/epoch - 2s/step
Epoch 2/2
13/13 - 8s - loss: 2.1338 - accuracy: 0.2116 - 8s/epoch - 591ms/step

Model ResNet50 4 GPUs
-----
Batch_size: 8192
Num_replicas: 4

Epoch 1/2
7/7 - 46s - loss: 2.3175 - accuracy: 0.1520 - 46s/epoch - 7s/step
Epoch 2/2
7/7 - 5s - loss: 2.1521 - accuracy: 0.2072 - 5s/epoch - 656ms/step
```

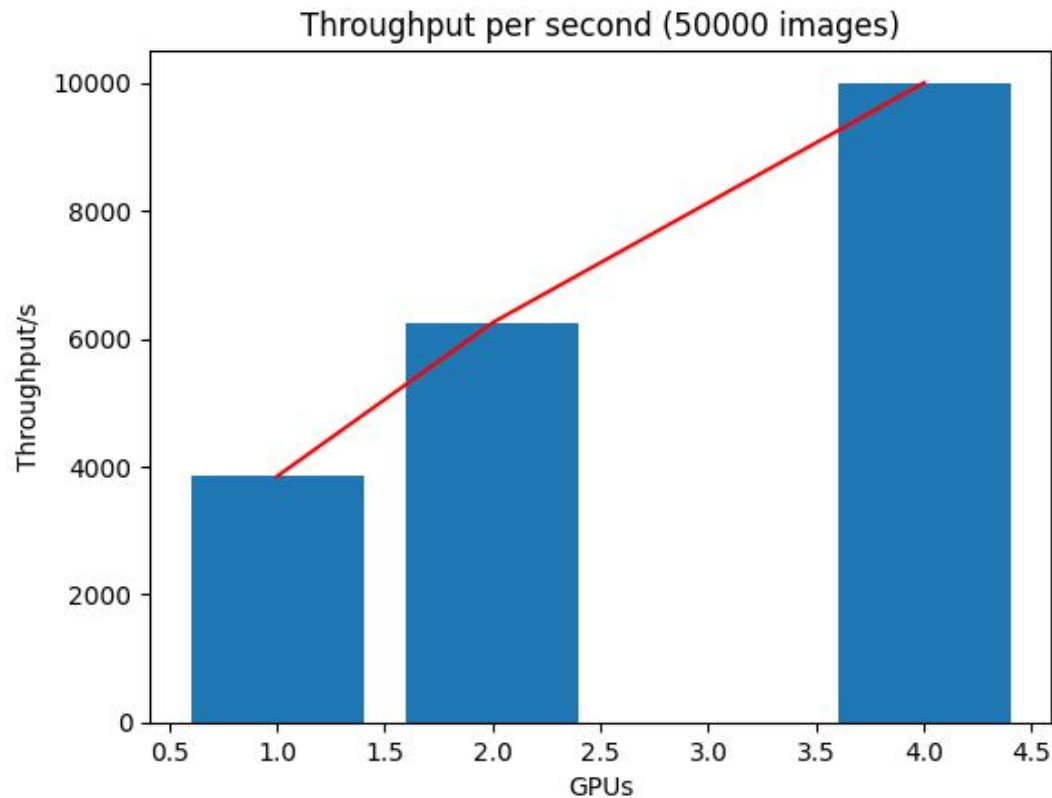
Task 10.6 Analyze the Impact of GPU Parallelism

- Sublinear evolution (red curve is convex)



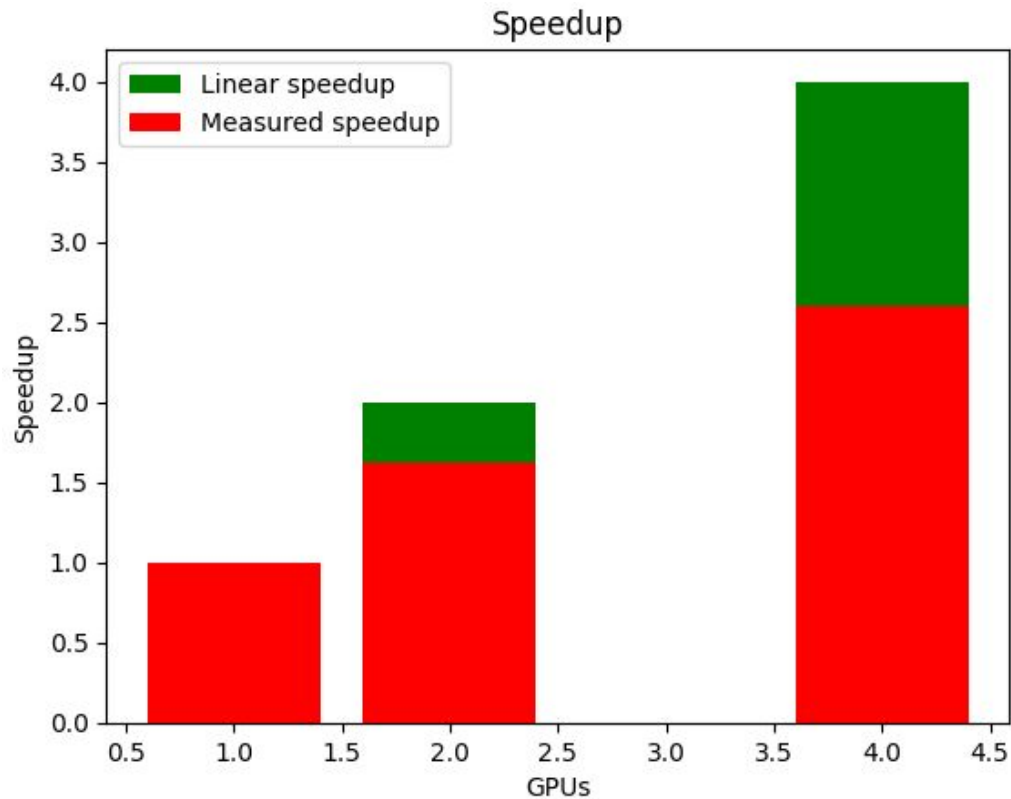
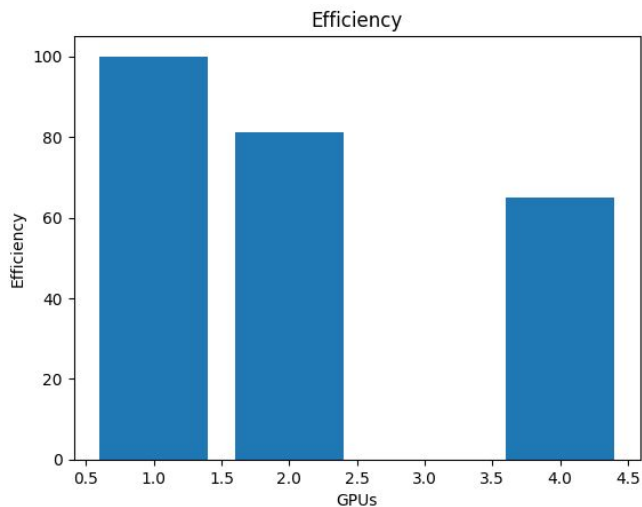
Task 10.6 Analyze the Impact of GPU Parallelism

- Quasi-linear evolution



Task 10.6 Analyze the Impact of GPU Parallelism

- Suboptimal speedup



Task 10.7 Parallelization of ResNet152

```
Model ResNet152 1 GPUs
```

```
-----  
Batch_size: 1024
```

```
Num_replicas: 1
```

```
Epoch 1/2
```

```
49/49 - 62s - loss: 2.2693 - accuracy: 0.1450 - 62s/epoch - 1s/step
```

```
Epoch 2/2
```

```
49/49 - 32s - loss: 2.1495 - accuracy: 0.2063 - 32s/epoch - 663ms/step
```

```
Model ResNet152 2 GPUs
```

```
-----  
Batch_size: 2048
```

```
Num_replicas: 2
```

```
Epoch 1/2
```

```
25/25 - 75s - loss: 2.2724 - accuracy: 0.1445 - 75s/epoch - 3s/step
```

```
Epoch 2/2
```

```
25/25 - 18s - loss: 2.1868 - accuracy: 0.1925 - 18s/epoch - 703ms/step
```

```
Model ResNet152 4 GPUs
```

```
-----  
Batch_size: 4096
```

```
Num_replicas: 4
```

```
Epoch 1/2
```

```
13/13 - 127s - loss: 2.3295 - accuracy: 0.1448 - 127s/epoch - 10s/step
```

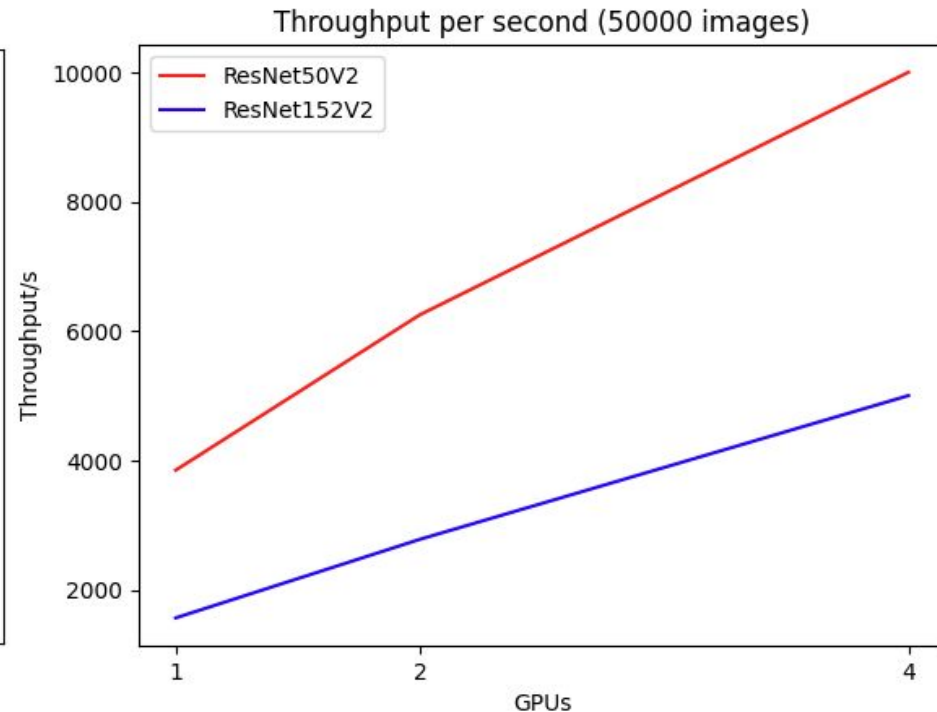
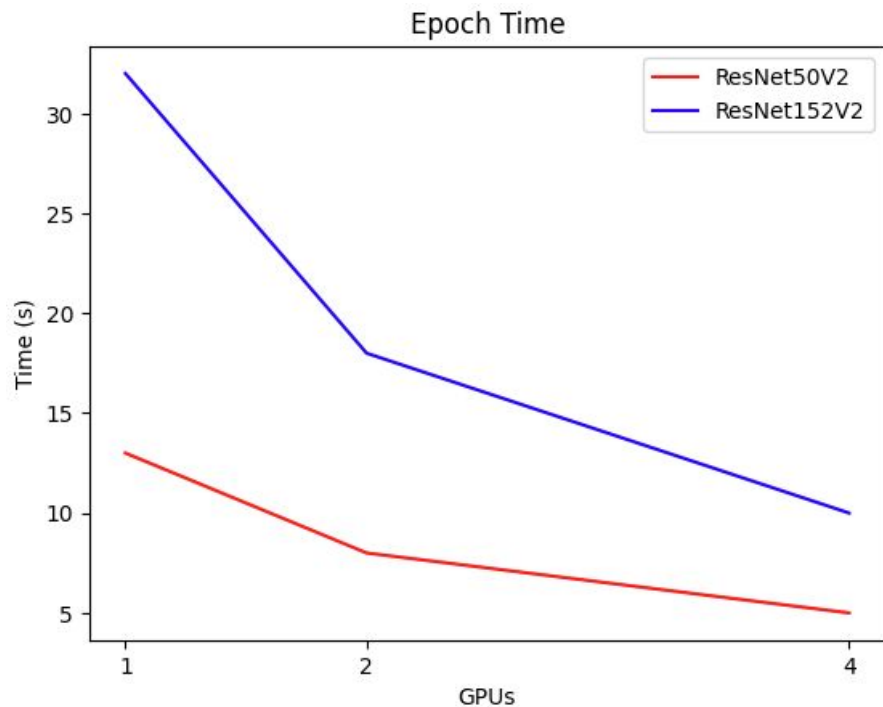
```
Epoch 2/2
```

```
13/13 - 10s - loss: 2.1791 - accuracy: 0.1923 - 10s/epoch - 785ms/step
```

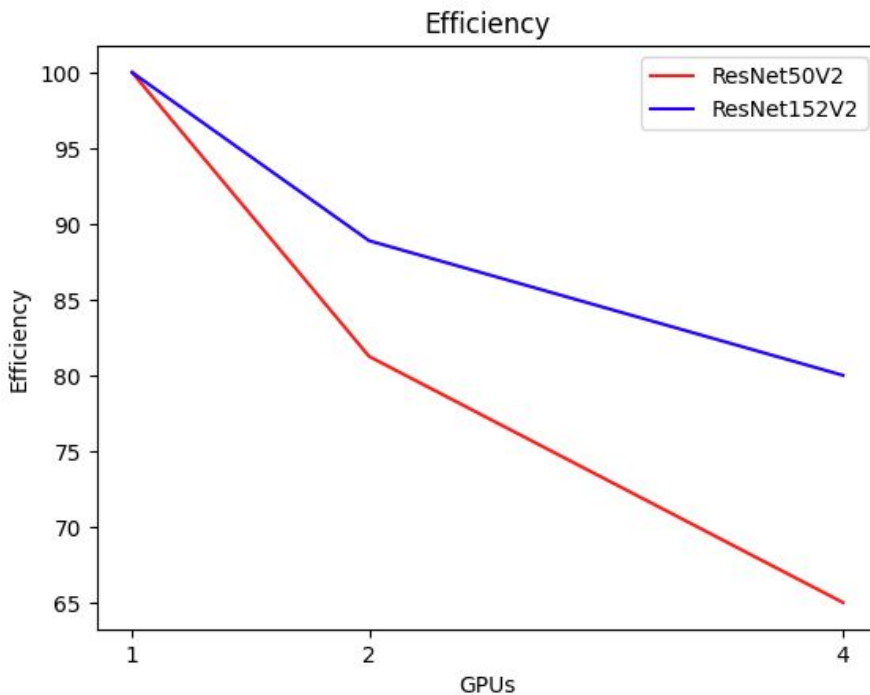
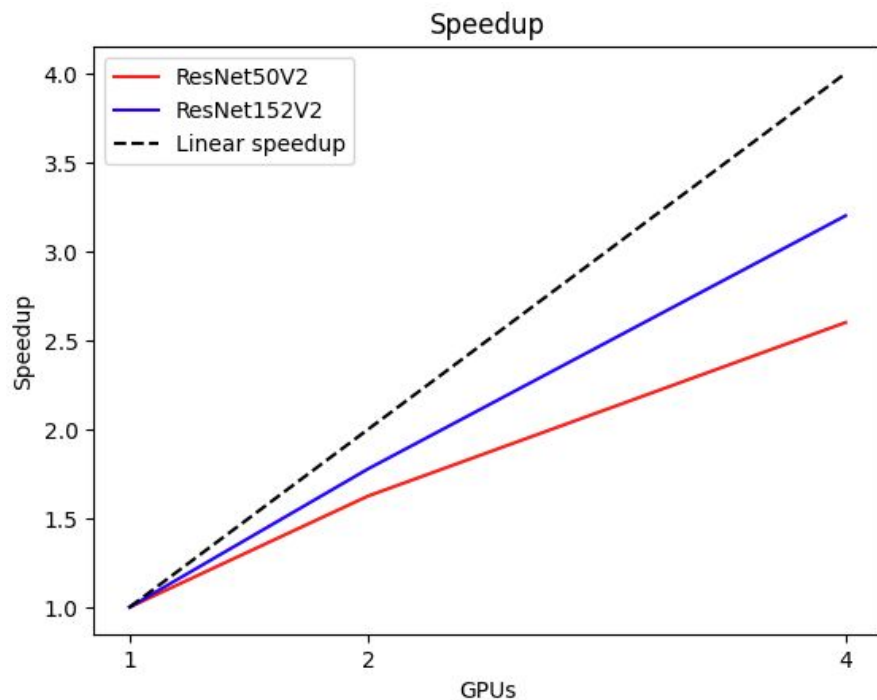
- Note:

Because of OOM error, we had to lower batch sizes by half compared to experiments with ResNet50.

Task 10.8 Comparing Parallel Training Performance Across Model Sizes



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- Why is the epoch time for ResNet152 significantly higher than for ResNet50?

ResNet152 has many more layers and parameters.

- Why does ResNet152 achieve lower throughput even with more GPUs?

Every image costs more computations, more GPUs help, but they must do more work per image, so throughput is lower.

- How do speedup and efficiency curves differ between the two models?

ResNet152 shows better speedup and efficiency for higher number of GPUs (closer to ideal scaling).

- Is the speedup less consistent for larger models? Why?

In our experiment it's closer to linear speedup, but different per-GPU memory pressure may make runs noisier, so measured speedup may vary more between runs.

- Could the deeper architecture of ResNet152 lead to more synchronization overhead?

Yes - it can cause out of memory issue and it increases parameter synchronization cost.

Task 10.9 ResNet101V2 Scalability Analysis

Model ResNet101 1 GPUs

Batch_size: 1024

Num_replicas: 1

Epoch 1/2

49/49 - 44s - loss: 2.2599 - accuracy: 0.1573 - 44s/epoch - 897ms/step

Epoch 2/2

49/49 - 22s - loss: 2.1369 - accuracy: 0.2200 - 22s/epoch - 457ms/step

Model ResNet101 2 GPUs

Batch_size: 2048

Num_replicas: 2

Epoch 1/2

25/25 - 52s - loss: 2.2437 - accuracy: 0.1635 - 52s/epoch - 2s/step

Epoch 2/2

25/25 - 14s - loss: 2.1023 - accuracy: 0.2276 - 14s/epoch - 566ms/step

Model ResNet101 4 GPUs

Batch_size: 4096

Num_replicas: 4

Epoch 1/2

13/13 - 83s - loss: 2.2897 - accuracy: 0.1467 - 83s/epoch - 6s/step

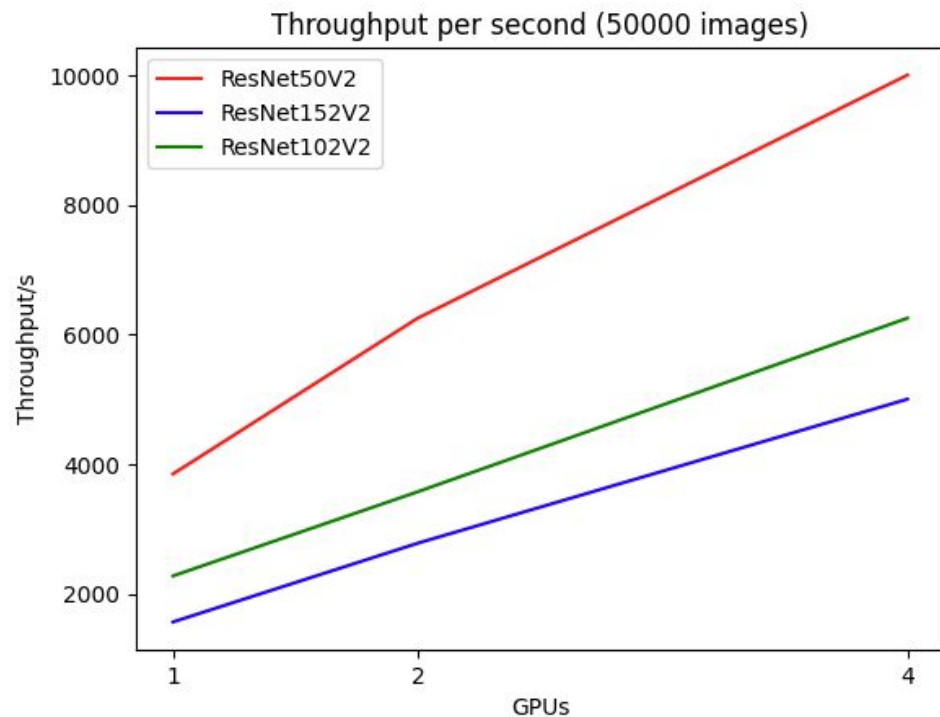
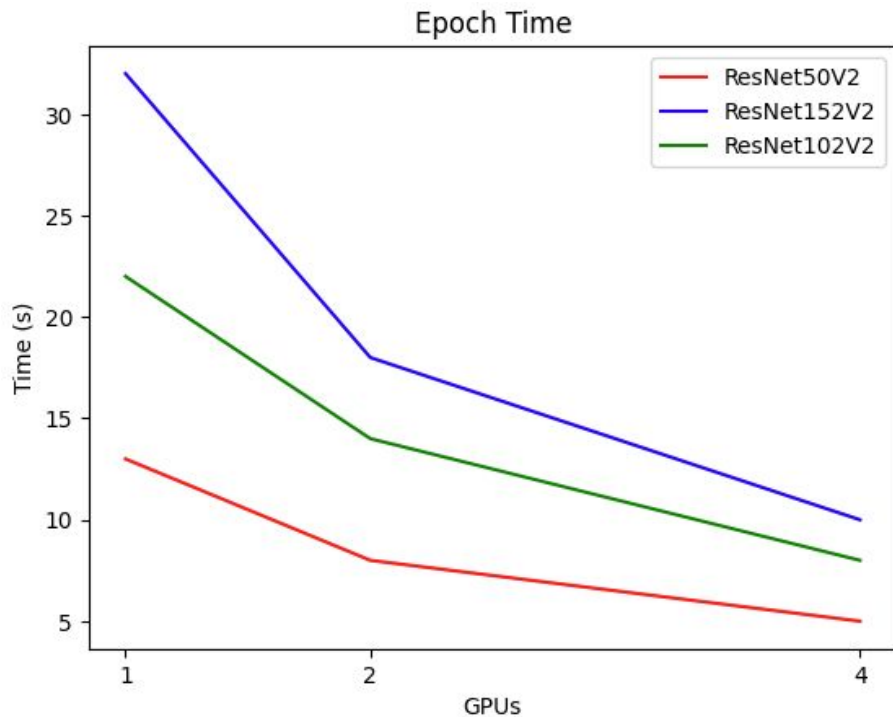
Epoch 2/2

13/13 - 8s - loss: 2.1529 - accuracy: 0.2084 - 8s/epoch - 579ms/step

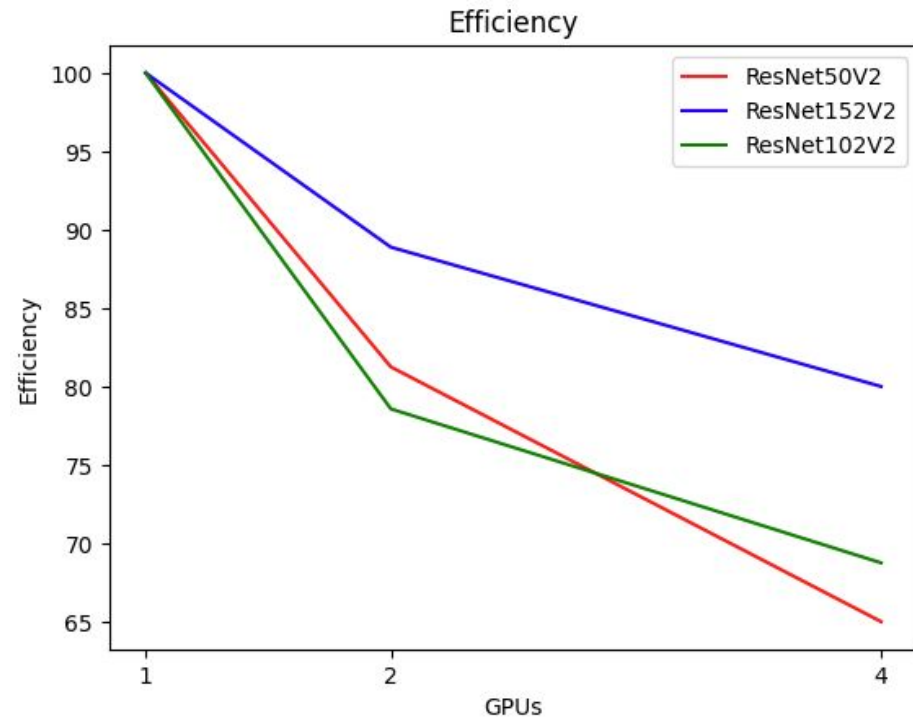
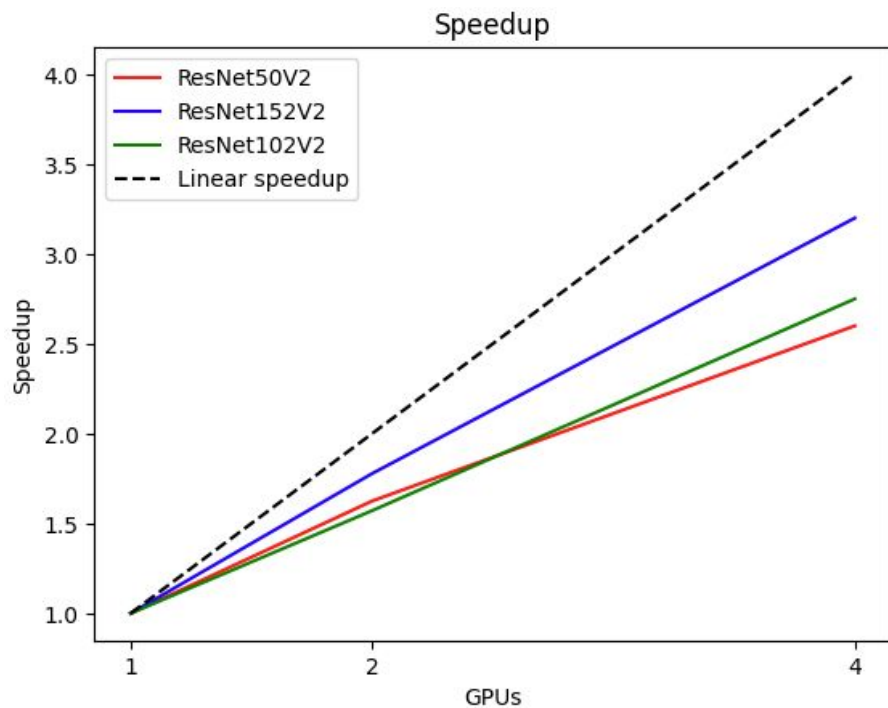
- Note:

Batch sizes are same as in ResNet152 experiments, 2 times lower than in ResNet50 experiments.

Task 10.9 ResNet101V2 Scalability Analysis



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Task 10.9 ResNet101V2 Scalability Analysis

- Does ResNet101V2 scale more like ResNet50 or ResNet152?

Speedup and efficiency of ResNet101 are much closer to results of ResNet50, but the overall times are between the other two.

- How does model size impact throughput, speedup, and efficiency?

The bigger the model, the lower the throughput. However, bigger models have better scores of speedup and efficiency for higher number of GPUs, closer to ideal scaling.

- Where do you observe diminishing returns or increased overhead?

Overhead increase with the size of ResNet, and is not worth the cost for those batch size. Its better scaling could show that the model will perform better for even larger CPU number.